

# Xiyuan GUO

tommyguo024@outlook.com — (516) 474-3279 — xguo.info — github.com/xguot

## EDUCATION

University of Virginia

Charlottesville, VA — Sept 2024 – May 2028 (Expected)

Degree: Bachelor of Arts in Computer Science and Mathematics — GPA: 3.89/4.00

Relevant Coursework: Quantum Mechanics, Linear Algebra, Data Structures & Algorithms

## PROFESSIONAL EXPERIENCE

Research Software Engineer, UVA (Ke's Group)

Charlottesville, VA — Oct 2025 – Present

- Orchestrated the end-to-end deployment of a highly available full-stack research platform on RHEL virtual machines, independently configuring all server-side web infrastructure, robust network security protocols, and relational database storage architectures.
- Applied quantum mechanics principles to engineer complex Density Functional Theory (DFT) reconstruction algorithms, bridging theoretical atomic modeling with robust software implementations to accurately catalog rare earth element electronic material properties.
- Engineered a cloud-native machine learning pipeline utilizing Go and Next.js, enabling global academic collaborators to securely share specialized material datasets and execute pre-trained deep predictive models via an intuitive interactive web interface.
- Created immersive, browser-based data visualizations using Three.js and JSmol, generating dynamic interactive 3D atomic models and Kiviat diagrams to facilitate the real-time structural inspection and validation of complex theoretical computational modeling outputs.

Machine Learning Assistant, Yale (Liu's Group)

New Haven, CT — Feb 2026 – Present

- Formulated a deep unrolled ADMM framework to solve ill-posed inverse problems in computational imaging, reconstructing multi-dimensional data encompassing spatial, spectral, and full-Stokes polarization domains across vast parameter spaces exceeding  $10^6$  dimensions.
- Engineered a specialized PyTorch pipeline utilizing a custom convolutional U-Net architecture as a learned proximal operator, specifically designed to mitigate non-Gaussian hardware sensor noise and yielding a 4.5 dB increase in PSNR over standard computational baselines.
- Implemented rigorous physics-informed constraints within custom deep learning model optimization loops, mathematically grounding complex vectorial Stokes parameter representations utilizing complex numbers, trigonometric formulations, and 3D Poincaré sphere mappings.
- Evaluated the architectural efficacy of novel Convolutional Physics-Informed Neural Networks against optical datasets, demonstrating that strictly constraining the multi-dimensional loss optimization landscape reduces spurious background image artifacting by well over 35%.

Quantitative Researcher, UVA (BAL Lab)

Charlottesville, VA — Apr 2026 – Present

- Engineered a mathematical framework for multivariate longitudinal missing data imputation that enforces the Law of Total Variation, mitigating severe structural covariance bias inherent in unconstrained predictive machine learning models (missForest, missRanger, mice).
- Formulated a novel Lagrange-constrained optimization engine, deriving custom matrix calculus gradients from first principles to mathematically project non-parametric Random Forest estimates onto a strictly defined smooth Riemannian positive-semidefinite (PSD) manifold.
- Architected a custom high-performance C++17 gradient descent solver via RcppArmadillo to iteratively evaluate dual Lagrangian multipliers, ensuring the final global variance of the imputed dataset converges to strict theoretical asymptotic population statistical expectations.
- Designed a Manifold Mapping pipeline utilizing robust statistical estimators (Spearman and MAD) to bound the highly non-convex search optimization space, scaling dense matrix operations across Slurm HPC clusters for massive Monte Carlo full-grid simulations.

Network Science Intern, UVA Biocomplexity Institute

Charlottesville, VA — May 2026 – Present

- Formulated massive-scale network simulations to dynamically model the stochastic spread of viral pathogens (e.g., COVID-19 and Ebola), implementing SEIR compartmental cascade models to capture heterogeneous transmission dynamics across global populations.
- Engineered attention-based Graph Neural Network (GNN) architectures to predict multi-hop epidemic propagation pathways, leveraging spatiotemporal message-passing frameworks to extract predictive latent topological features from large-scale dynamic contact networks.
- Collaborated within a transdisciplinary initiative to scale core epidemiological algorithms, bridging high-performance Bayesian data science methodologies with empirical public health parameters to inform and optimize proactive global disaster resilience strategies.

## AWARDS & HONORS

- Double Hoo Research Award:** Secured competitive grant funding to drive independent, interdisciplinary computational research.
- Academic Honors:** USA Computing Olympiad (USACO) Silver Division; UVA Dean's List of Distinguished Students.
- Open-Source Recognition:** Achieved Hacker News Front Page (#5, #6), Homebrew Core inclusion, and 342+ GitHub stars for `difi`.

## CERTIFICATIONS & SKILLS

- Math & Science:** Quantum Mechanics, Density Functional Theory (DFT), Linear & Tensor Algebra, Real Analysis, ODEs
- Machine Learning & AI:** PyTorch, TensorFlow, Keras, JAX, Scikit-Learn, XGBoost, Pandas, NumPy, OpenCV, RcppArmadillo
- DevOps & Cloud:** Linux (RHEL/Fedora), AWS (EC2, S3), Docker, Kubernetes, CUDA, Slurm HPC, Git, CI/CD, Supabase, Sway, Nvim
- Frameworks & Web:** React, Next.js, SvelteKit, Node.js, FastAPI, Bubble Tea, GraphQL, REST APIs, TailwindCSS
- Languages:** Python, C/C++, Rust, Go, TypeScript, JavaScript, R, Julia, SQL, Lua, Bash, Java, L<sup>A</sup>T<sub>E</sub>X
- Certifications:** AWS Certified AI Practitioner, AWS Certified Cloud Practitioner